a What a case contains (case BACT-MEN-RS02)

Shown to the agent at the start | Returned only if ordered | Hidden; used for scoring

Presentation

"Severe headache, high fever and progressive confusion over 18 hours."

Man, 38, emergency department; VP shunt since infancy.

T 39.8 °C	HR 112
BP 142/88	SpO_2 97%

Nuchal rigidity; Kernig and Brudzinski signs; GCS 9 (E2 V3 M4).

532 narrative words (median)

Initial results

WBC	22.8 ×10⁹/L ↑
CRP	218 mg/L ↑
Procalcitonin	8.4 ng/mL ↑
Sodium	131 mEq/L ↓
Lactate	3.2 mmol/L ↑
Neutrophils	83% ↑
D-dimer	2.8 µg/mL ↑
INR	1.2 ↑

Studies already done when the agent first sees the patient.

4.0 stored studies per case (mean)

Ordered results

`analyze_csf`
Turbid; WBC 3,200/µL; opening pressure 32 cmH_2O

`order_microbiology`
S. pneumoniae in both bottles at 14 h

`analyze_brain_mri`
Diffuse leptomeningeal enhancement

Unstored orders return a shared normal report.

9.6 stored results per case (mean)

Policy

required actions	3.8
recommended	1.6
actions to avoid	5.7
order constraints	1.5
accepted labels	7.3
vetoed labels	11.0

items per case (mean), plus a stop rule

b 20 conditions, 30 cases each

Cerebrovascular	Ischaemic stroke
	Subarachnoid haemorrhage
Infection and inflammation	Bacterial meningitis
	Anti-NMDAR encephalitis
	Multiple sclerosis
Neurodegeneration	Early Alzheimer's disease
	Frontotemporal dementia
	Vascular dementia
	Normal-pressure hydroc.
	Parkinson's disease
	ALS
Seizures	Temporal lobe epilepsy
	Status epilepticus
Neuromuscular	Guillain–Barré syndrome
	Myasthenia gravis
Oncology	High-grade glioma
Headache	Migraine with aura
Functional	Functional neurol. disorder
Systemic mimics	Cardiac syncope
	Hepatic encephalopathy

train **500** · test, seen condition **40** · test, held out **60**

Difficulty: simple 153 · moderate 238 · puzzle 209

Setting: ED 240 · inpatient 180 · outpatient 180

c Physician findings are fixed in every case they affect

1 Generate
Condition-expert LLM agents draft cases and policies from guideline criteria (WHO, NICE, AHA, IDSA, ILAE, EAN, AAN).
20 criteria packs

→

2 Validate
Schema, consistency and realism checks; an agent that follows the policy must receive every required result.
600 cases pass

→

3 Physician review
Blind review of seven dimensions per case by two independent physicians; a third on disagreement.
20 condition review rounds

→

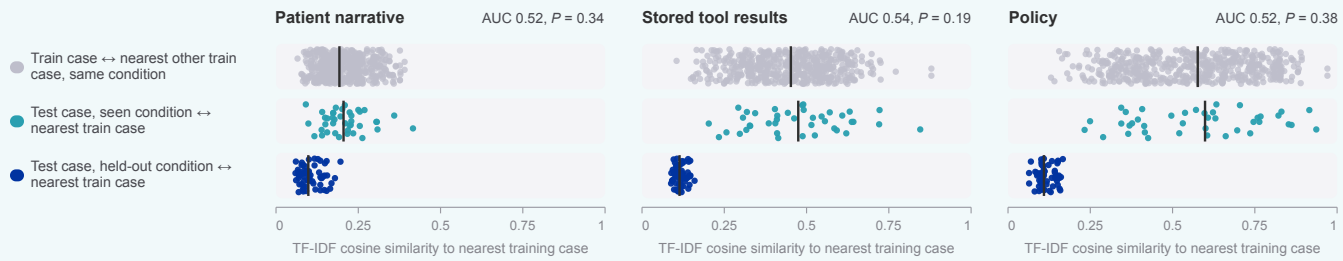
4 Fix the class
Each finding is treated as a defect class and corrected in every case it affects, not only the one reviewed.
26 dataset-wide revisions

→

5 Approve
A policy becomes reportable only after two independent approvals on all seven dimensions.
draft → approved

every fix is re-validated on all 600 cases

d Test cases are no closer to training than training cases are to one another



Dots, cases; black bars, medians. AUC and one-sided Mann–Whitney *P*: test cases of seen conditions against the train ↔ train baseline (AUC 0.5, no closer than chance).